A Real-Time Framework for Assessing Patient Pain and Discomfort from Facial Expressions Using Deep and Classical Machine Learning Approaches

Halak V. Dudiya, P. Thulasi Ramani, Soumar Sikhar Talukdar, Mahipal Amrutlal Charaniya
Department of Computer Engineering
Marwadi University
Rajkot, Gujarat, India – 360003

***Abstract*—During a physiotherapy session, the way a patient's face moves often says more about their tolerance and engagement than anything they say aloud, yet clinicians rarely have the time to watch for these signals continuously. Manual observation is subjective by nature, and fleeting micro-expressions of discomfort are easy to miss when the therapist's attention is split between coaching and watching. This work develops a real-time facial analysis pipeline that flags pain and discomfort as they happen, relying on facial landmarks tracked from live video and a set of compact machine learning classifiers. Landmark coordinates are extracted frame by frame, normalized into a structured feature set, and stored as a labeled CSV dataset recorded from live subjects. Several classifiers, including a Multi-Layer Perceptron (MLP), a Support Vector Machine (SVM), a Random Forest, and a neural network variant, were trained and compared for live inference. The strongest configuration reached high accuracy together with solid precision, recall, and F1-score over the pain and non-pain classes, suggesting that the pipeline is dependable enough to assist physiotherapists in monitoring patient discomfort throughout rehabilitation.**

***Index Terms*—facial landmarks, real-time detection, pain expression, machine learning, computer vision**

I. Introduction

Facial Expression Recognition (FER) reads the movements of a face to infer emotion, behavior, or physical state, and it has found a natural home in healthcare, safety monitoring, ergonomics, and human–computer interaction. Within clinical settings such as physiotherapy, catching a patient's discomfort early matters: pain expressions surface for only a moment, and a therapist juggling instruction and supervision can easily overlook them. An automated detector that watches continuously can prompt more responsive care, lower the chance of aggravating an injury, and generally improve how treatment feels for the patient.

Pain expression analysis has been studied in several clinical contexts, from postoperative monitoring to assessment during procedures and remote health surveillance. Machine learning has proven capable of picking out pain-related facial cues through both appearance-based and landmark-based representations [1], [2]. FER models have likewise been applied to ergonomics, where strain during repetitive work can be identified from the face [3], and to safety-oriented real-time monitoring [4]. Even so, most published systems assume large, well-curated datasets or controlled laboratory conditions, which limits how readily they transfer to live physiotherapy, where illumination, head pose, and person-specific expression habits all vary.

Real-time operation adds its own difficulties: constrained compute, scarce training data, and strong inter-person variability in how pain shows on the face. Many models that shine on static benchmarks degrade when fed a live video stream, and small datasets collected from few subjects make stable feature design essential rather than optional.

To tackle these issues, we built a lightweight, practical pain-expression recognition system aimed specifically at physiotherapy. The contributions are:

1) A real-time facial landmark dataset, saved in CSV form, covering pain and neutral expressions recorded from one or more subjects.
2) Training of a pain-expression classifier with several machine learning models, namely an MLP, an SVM, and a Random Forest.
3) A real-time detection pipeline that pulls landmarks from a live webcam feed and classifies each frame on the fly.
4) A comparative study of three algorithms over accuracy, precision, recall, and F1-score to select the model best suited to live deployment.

By pairing efficient landmark extraction with both classical and deep learning classifiers, the resulting system offers a flexible, deployable route to real-time pain monitoring in physiotherapy and, potentially, other healthcare contexts.

II. Related Work

Therapists commonly struggle to follow a patient's pain or emotional state while the patient exercises, and self-report based monitoring introduces bias that differs from one patient to the next, complicating any attempt to adjust therapy accordingly.

Zhang et al. [3] surveyed how artificial intelligence can support pain assessment, finding that machine learning and natural language processing perform reasonably well at recognizing pain and patient emotion. Their review also exposed a persistent weakness: most efforts remain pilot studies that were never trialed in real clinics with heterogeneous patient populations. Gwak and Park [4] approached the user-experience side of the problem by building a face rehabilitation

app and testing design variations in shape and transparency. Preferences turned out to be age-dependent, with older users favoring opaque feedback pages and younger users preferring translucent ones. Their evaluation, however, covered only simple expressions such as smiling rather than the complex action units that actually characterize pain. On the systems side, Ciraolo et al. [2] targeted the computational bottleneck of tele-rehabilitation, constructing a FER module that runs on low-resource edge devices. Their pipeline, like ours, combines MediaPipe-derived facial features with lightweight classifiers such as SVM and Random Forest, and it struck a reasonable speed–accuracy balance, though it was never validated on genuinely low-power hardware and its accuracy remained below what dependable clinical use would demand. Related efforts have explored machine-learning pose estimation as a digital biomarker for motor rehabilitation delivered remotely [5] and full tele-rehabilitation-as-a-service platforms [6], reinforcing that lightweight, service-oriented analysis is an active engineering goal. Taking a broader view, Antel et al. [1] systematically reviewed 197 articles on AI in pain management and confirmed that while diagnosis and prediction are well covered, validation across diverse clinical settings and patient groups is scarce, leaving a visible gap between research output and bedside practice. Our work responds to these shortcomings by adopting a lightweight design in the spirit of [2], focusing on the physiotherapy context that [4] left unexplored, and validating on a purpose-built dataset that goes beyond a pilot-scale study in the sense criticized by [3] and [1]. Computer vision monitoring tailored to home-based physiotherapy has also been reviewed recently [7], confirming growing interest in exactly this deployment scenario.

Beyond the pain literature, FER has advanced rapidly over the past decade on the strength of deep networks and large annotated corpora, as summarized in recent reviews [8]. Earlier fusion-based recognition schemes date back well before the deep learning era [9]. CNN approaches perform impressively on benchmarks such as FER2013, AffectNet, and CK+, which together span tens of thousands of labeled images of basic emotions [8]. The catch is that these models demand heavy computation and large training sets, which conflicts with lightweight, real-time deployment.

Landmark-based methods take a different route, modeling facial geometry rather than raw texture. Predictors such as Dlib's 68-point model and Google's MediaPipe Face Mesh, which tracks 468 landmarks, yield structured facial data at very low cost; extraction is fast, tolerates moderate lighting changes, and does not require massive labeled corpora, which is why such representations dominate real-time pipelines. Lightweight deep architectures have pushed in the same direction, achieving high-speed emotion recognition on modest hardware [10].

Pain recognition itself has matured into a specialized FER branch, surveyed in detail by Pini et al. [11]. The UNBC-McMaster Shoulder Pain Expression Archive is the canonical dataset in this area: patients perform shoulder movements while experiencing graded discomfort, and most studies on it apply appearance features or deep spatio-temporal networks to estimate pain intensity [12]. Landmark-driven variants have likewise been proposed, fusing deep models over facial landmark features for pain level prediction [13], and compact CNNs have been shown capable of estimating pain intensity from expressions alone [14]. Micro-expression oriented approaches that combine transfer learning with deep CNNs have also been reported [15]. These methods score well but typically assume high-resolution video, large training sets, and expensive architectures, which is a poor fit for physiotherapy clinics where real-time response is the priority.

Real-time FER frameworks have meanwhile spread into driver monitoring, engagement detection, and human–robot interaction. Systems like OpenFace and MediaPipe demonstrate that real-time recognition is achievable without deep CNNs by combining landmark extraction with simple classifiers and efficient feature encodings.

Building on these threads, the system proposed here prioritizes simplicity, efficiency, and live usability: instead of a heavy CNN trained on big corpora, facial landmarks feed compact classifiers (MLP, SVM, Random Forest). This keeps the computational footprint small while preserving accuracy, and it runs on ordinary laptop-plus-webcam hardware, which suits physiotherapy settings where GPUs are rarely available.

III. Methodology

This section lays out the complete architectural pipeline for real-time facial monitoring and pain intensity analysis. The design follows a three-stage cascade intended to keep expression analysis dependable even in uncontrolled physiotherapy environments.

A. System Architecture

The pipeline quantifies pain and discomfort objectively during active therapy, which demands both high-fidelity expression analysis and robustness to environmental noise. As sketched in Fig. 1, the workload is decomposed into sequential, interdependent sub-problems. This decomposition improves computational efficiency through conditional decision-making, so that intensive processing runs only when genuine distress signals appear. The input is a video stream V made up of time-indexed frames,

$$F = \{F_1, F_2, F_3, \ldots, F_T\}, \tag{1}$$

with early-termination logic built around the patient's recovery state.

B. Data Collection

A webcam-driven collection procedure was adopted: each captured frame had its landmark coordinates extracted and written to a CSV file together with a class label. Expression categories included Happy, Neutral, Angry, and Pain, giving N labeled samples after encoding all classes.

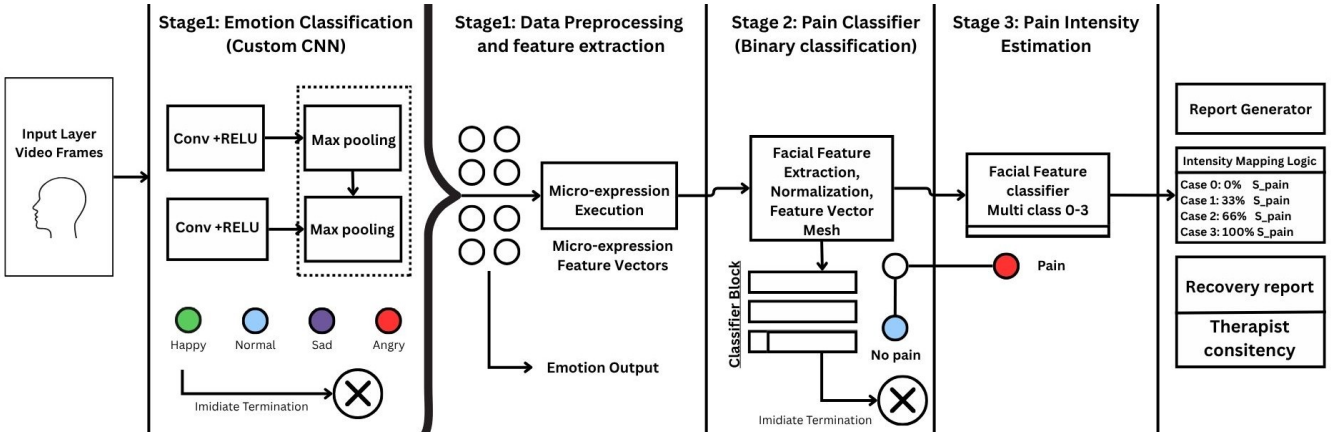


Fig. 1. Overall system architecture showing the three-stage pipeline: emotion classification, pain classification, and pain intensity estimation.

C. System Overview and Sequential Processing

The system operates as a continuous data pipeline that ingests video and splits it into frames. Three discrete models cooperate: a Stage 1 emotion classifier, a Stage 2 binary pain detector, and a Stage 3 pain intensity estimator that maps expression evidence onto graded severity levels.

1) Input Pipeline and Preprocessing: Processing begins at the gateway, which accepts raw video and applies the foundational preprocessing shared by all downstream models.

- **Facial detection and alignment:** detectors such as MTCNN locate and track the patient's face reliably across consecutive frames.
- **Landmark tracking and normalization:** high-precision landmark detection identifies key fiducial points, which are then normalized to cancel scale, rotation, and translation drift caused by natural movement during exercise.
- **Data filtering:** morphological operations remove residual noise, producing cleaner and more continuous contours suited to analysis.

2) Real-Time Operation and Adaptive Termination: Flow control is adaptive and driven by Stage 1 output. If a "Happy" expression persists beyond a set window during therapy, the pipeline routes to its terminal state, ending the session log and emitting a healthy status report. This both trims computational cost and gives the therapist a positive signal.

D. Feature Engineering and Data Representation

1) Dual Extraction Strategy: Two feature families are produced:

- **Appearance features:** normalized image patches (225×225 RGB or 48×48 grayscale) that serve as input to the deep convolutional network.
- **Geometric features (F_V, Stages 2 and 3 input):** statistics derived from normalized landmark displacement and configuration, capturing localized deformations that map onto the action units associated with pain.

2) Geometric Feature Vector: The feature vector F_V consumed by Stages 2 and 3 is engineered around the clinically validated core pain action units: brow lowering (AU4), orbital tightening (AU6 and AU7), levator contraction (AU9 and AU10), and eye closure (AU43), following the mapping established in the facial pain analysis literature [11], [13]. Concrete measurements include `mouth_open`, `eye_open`, `brow_eye`, and `nose_center_offset`. For example, `brow_eye` encodes the vertical narrowing tied to AU4, while `eye_open` supports detection of AU6/AU7 orbital tightening and AU43 eye closure. Relevant action units are illustrated in Fig. 2. Grounding the input space in these clinically derived proxies keeps pain quantification interpretable and scientifically anchored.

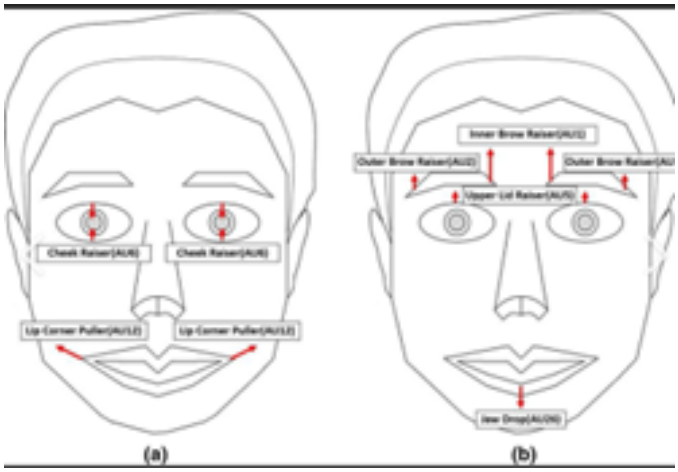


Fig. 2. Facial action units related to pain analysis. (a) Cheek raiser (AU6) and lip corner puller (AU12). (b) Brow raisers (AU1, AU2), upper lid raiser (AU5), and jaw drop (AU26).

E. Stage 1: Custom CNN Emotion Classification Model

1) CNN Architecture: The emotion classifier uses a custom CNN tuned for real-time FER, following the general principles that made deep convolutional models effective for image classification [16].

Algorithm 1 Real-time facial expression recognition

Require: Input image I, trained model $\mathcal{M}$, classifier $\mathcal{C}$
Ensure: Annotated image I_{out}

```
I_out ← I
G ← RGB2Gray(I)
F ← DetectFaces(G, C)
if F = ∅ then
    return I_out
end if
for each (x, y, w, h) ∈ F do
    R ← G[y : y + h, x : x + w]
    R' ← Resize(R, 48 × 48)
    R_norm ← R'/255.0
    P ← M(R_norm)
    L ← argmax(P)
    I_out ← Draw(I_out, L, (x, y, w, h))
end for
return I_out
```

- **Input layer:** accepts normalized images (e.g., 225×225×3 RGB or grayscale).
- **Convolutional layers:** at least three primary convolutional layers with ReLU activations; early layers apply filters (e.g., 5×5 kernels) at a stride chosen for efficient feature map generation.
- **Pooling layers:** two max-pooling stages (e.g., 3×3 windows) reduce dimensionality and add invariance to small shifts.
- **Output layers:** fully connected layers feeding a softmax that assigns each frame to one of the affective states Happy, Neutral, Sad, or Angry.

Algorithm 1 summarizes the per-frame inference loop.

2) Training Protocol: The CNN is trained on large, diverse labeled corpora such as FER2013 and the Extended Cohn–Kanade (CK+) dataset, both widely used FER benchmarks [8]. Transfer learning strengthens generalization, and augmentation counters the class imbalance typical of FER data. After training, weights are serialized (typically in HDF5) for efficient deployment and loading.

F. Stage 2: Binary Pain Classifier

1) Feature Augmentation for Micro-Expression Detection: Stage 2 performs the binary task C_{pain} (pain vs. no pain) over the geometric vector F_V. Because pain during exercise often appears as brief micro-expressions, F_V is augmented with temporal deltas ΔF_V that encode the velocity and rate of change of landmark displacement between consecutive frames, letting the classifier catch the rapid onset and offset typical of subtle pain-related action unit activity, a strategy consistent with micro-expression based pain analysis [11], [15].

2) Model Selection and Objective: A feature-driven classifier such as an SVM or a light MLP is trained on the combined static vector F_V and temporal vector ΔF_V. The objective is tuned toward high recall on the "Pain" class: in clinical safety monitoring, a false negative (missed real pain) is the costly error, so recall is prioritized even at some expense in precision. Frames classified as no-pain trigger a secondary termination T_{NP} to conserve processing. Algorithm 2 details the MobileNetV2 transfer learning procedure used for this stage.

Algorithm 2 MobileNetV2 transfer learning for pain classification

Require: Dataset D, pre-trained MobileNetV2 θ_{base}
Ensure: Optimized model $\mathcal{M}^*$

1: **Preprocessing:** resize $x \to 128\times128$, normalize to $[-1, 1]$
2: **Augmentation:** $\Phi(x) \leftarrow$ Rotate, Shift, Zoom, Flip(x)
3: **Initialize:** freeze θ_{base}; add head GAP $\to$ BN $\to$ Drop(0.4) $\to$ FC(128) $\to$ BN $\to$ Drop(0.4) $\to$ Softmax
4: **Stage 1 (warm-up):** compile with Adam($lr = 1e^{-3}$); train head 8 epochs on $\Phi(D_{train})$
5: **Stage 2 (fine-tuning):** unfreeze last 30 layers of θ_{base}; compile with Adam($lr = 1e^{-4}$); train 12 epochs with EarlyStopping and ReduceLROnPlateau; keep best θ^* by validation loss
6: **return** $\mathcal{M}^*$

G. Stage 3: Random Forest Pain Intensity Model

1) Pain Level Classification: Frames that pass Stage 2 move on to intensity estimation, handled by a Random Forest (RF) classifier chosen for its speed, robustness, and native feature-importance reporting on structured, engineered inputs such as F_V. The RF assigns each frame to one of four discrete grades: Case 0 (no pain), Case 1 (slight), Case 2 (mild), or Case 3 (intense).

2) Intensity Mapping to a Clinical Scale: The discrete output C_{pain} is converted into a quantitative score S_{pain} on a 0–100 scale so that it aligns directly with standard clinical instruments such as the Numerical Rating Scale (NRS):

$$S_{pain} = \begin{cases} 0, & C_{pain} = 0 \text{ (no pain)}, \\ 33, & C_{pain} = 1 \text{ (slight pain)}, \\ 66, & C_{pain} = 2 \text{ (mild pain)}, \\ 100, & C_{pain} = 3 \text{ (intense pain)}. \end{cases} \tag{2}$$

This discretization gives enough granularity to track longitudinal recovery while keeping the score immediately actionable for the therapist.

Algorithm 3 describes the end-to-end assessment and visualization loop, including the per-frame intensity computation

$$I_t = \left(\frac{\sum_{c=1}^{|C|} c \cdot P_t(c)}{|C|} \right) \times 100, \tag{3}$$

where $P_t(c)$ denotes the model's class probabilities at frame t.

H. Integrated System Output and Reporting

1) Recovery Report Generator: The report generator condenses frame-level traces into a recovery report whose purpose is longitudinal progress tracking. Aggregated temporal metrics include:

Algorithm 3 Automated pain assessment and visualization system

Require: Video stream $\mathcal{V}$, trained model $\mathcal{M}$, scaler $\mathcal{S}$, face detector $\mathcal{D}$

Ensure: Annotated video $\mathcal{V}_{\text{out}}$, analysis report $\mathcal{R}$

Initialize: $C \leftarrow \{0{:}\text{No Pain}, 1{:}\text{Mild}, 2{:}\text{Severe}\}$; load $\mathcal{M}, \mathcal{S}$; set threshold $\tau \leftarrow 0.5$; $H_{\text{level}} \leftarrow []$, $H_{\text{intensity}} \leftarrow []$
for each frame F_t in $\mathcal{V}$ **do**
  $F_{\text{rgb}} \leftarrow \text{ConvertToRGB}(F_t)$
  $L \leftarrow \mathcal{D}.\text{detect}(F_{\text{rgb}})$ ▷ 68 facial landmarks
  if $L = \emptyset$ **then**
    $H_{\text{intensity}}.\text{append}(\text{NaN})$; **continue**
  end if
  Feature extraction: $\text{IOD} \leftarrow \|L_{\text{left_eye}} - L_{\text{right_eye}}\|_2$
  $x_{\text{raw}} \leftarrow \left[\frac{\text{Mouth}_{\text{dist}}}{\text{IOD}}, \frac{\text{Eye}_{\text{open}}}{\text{IOD}}, \frac{\text{Brow}_{\text{dist}}}{\text{IOD}}, \text{Nose}_y\right]$
  Inference: $x_{\text{scaled}} \leftarrow \mathcal{S}.\text{transform}(x_{\text{raw}})$
  $P_t \leftarrow \mathcal{M}.\text{predict_proba}(x_{\text{scaled}})$; $\hat{y}_t \leftarrow \text{argmax}(P_t)$
  Quantification: compute I_t via Eq. (3)
  $H_{\text{level}}.\text{append}(\hat{y}_t)$; $H_{\text{intensity}}.\text{append}(I_t)$
  Visualize: draw intensity meter I_t and label $C[\hat{y}_t]$ on F_t; write to $\mathcal{V}_{\text{out}}$
end for
Post-processing: $H_{\text{intensity}} \leftarrow \text{InterpolateNaNs}(H_{\text{intensity}})$
$H_{\text{smooth}} \leftarrow \text{MovingAverage}(H_{\text{intensity}}, \text{window} = 2\text{s})$
$\mathcal{R} \leftarrow \text{GeneratePlots}(H_{\text{smooth}}, H_{\text{level}})$
return $\mathcal{V}_{\text{out}}, \mathcal{R}$

- **Peak pain score** $S_{pain,\max}$**:** the maximum intensity recorded.
- **Average pain score** $\overline{S}_{pain}$**:** the mean intensity across all analyzed frames.
- **Endurance time:** total duration with $S_{pain} \geq 66$ (the moderate pain threshold in Eq. (2)).
- **Positive engagement time:** total duration spent in the Happy or Neutral state per Stage 1 output.

2) Dataset Requirements and Validation: Validation draws on domain-specific corpora: Stage 1 uses general FER benchmarks (FER2013, CK+) [8], while Stages 2 and 3 rely on pain action unit annotated data such as the UNBC-McMaster Shoulder Pain Expression Archive [11], [12], keeping the evaluation clinically relevant to physiotherapy. Class imbalance, a recurring issue in pain detection, is mitigated with weighted loss functions so that the minority "Pain" class retains sensitivity.

3) Performance Metrics and System Benchmarking: Evaluation covers both classification quality and clinical utility:

- **Weighted F1-score:** the primary metric for the nociception models (Stages 2 and 3), ensuring robustness across imbalanced pain classes.
- **Inference latency:** measured to confirm real-time behavior of the full cascade.
- **Clinical correlation:** final validation benchmarks the S_{pain} output against human expert pain ratings, checking that the machine-learned score behaves as a meaningful clinical indicator, in line with the validation gaps highlighted in [1], [3].

IV. Experimental Setup

This section documents the hardware and software configuration, dataset preparation, preprocessing pipeline, and evaluation metrics used across the three stages of the framework.

A. Hardware and Software Configuration

Hardware:

- Intel or AMD multi-core CPU.
- NVIDIA GPU with CUDA support, e.g., RTX 3060 with 6–12 GB VRAM.
- 16 GB RAM.

Software:

- Python 3.10 or later.
- TensorFlow/Keras 2.x for the CNN and MobileNetV2 models.
- scikit-learn 1.x for the Stage 3 Random Forest.
- OpenCV for image preprocessing.
- Matplotlib and Seaborn for visualization.

B. Dataset Description

Three datasets were prepared, one per pipeline task.
Stage 1: Emotion Classification:

- Classes: angry, happy, neutral, sad.
- Image size: 48×48 grayscale.
- Split: 70% train, 20% validation, 10% test.

Stage 2: Pain vs. No Pain Detection:

- Classes: pain, no pain.
- Split: 70% train, 20% validation, 10% test.
- Derived from Stage 1 non-happy samples.

Stage 3: Pain Intensity Prediction:

- Classes: 0 (no pain), 1 (slight), 2 (mild), 3 (intense).
- Features extracted using CNN embeddings.
- Classifier: Random Forest hybrid model.

C. Preprocessing Pipeline

The following operations were applied across all stages:

- Grayscale conversion.
- Histogram normalization.
- Resizing to 48×48 for Stages 1 and 2.
- Augmentation: rotation up to 15 degrees, shifts up to 10–15%, shear and zoom factors of 0.15, horizontal flip.

In Stage 3, features were standardized with `StandardScaler`.

D. Evaluation Metrics

The metrics used across all stages were:

- Accuracy.
- Precision, recall, and F1-score (macro and weighted).
- Confusion matrix.
- Loss and accuracy curves across training epochs.
- Mean squared error for Stage 3.
- Log loss for the Stage 3 hybrid model.

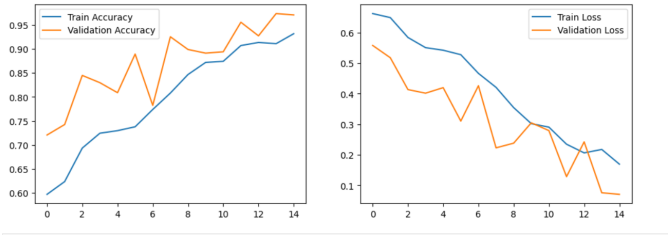


Fig. 3. Training and validation accuracy and loss for the Stage 1 CNN model.

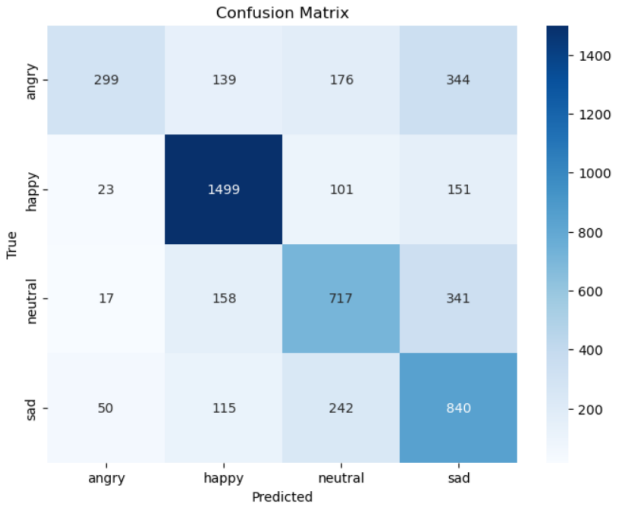


Fig. 4. Confusion matrix for the Stage 2 MobileNetV2 pain detector.

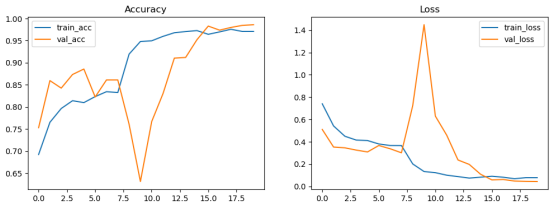


Fig. 5. Training and validation accuracy and loss for the Stage 2 MobileNetV2 classifier.

V. Performance Evaluation

This section compares the trained models, their convergence behavior, and their test set performance.

A. Model Comparison

1) Stage 1: Facial Emotion Classification: Table I reports Stage 1 outcomes, and Table II breaks performance down per class; the corresponding training behavior appears in Fig. 3.

TABLE I
Stage 1 Emotion Classification Results.

Model	Accuracy	Macro F1	Weighted F1
MobileNetV2	64.37%	0.60	0.63
Custom CNN	93.75%	0.94	0.94

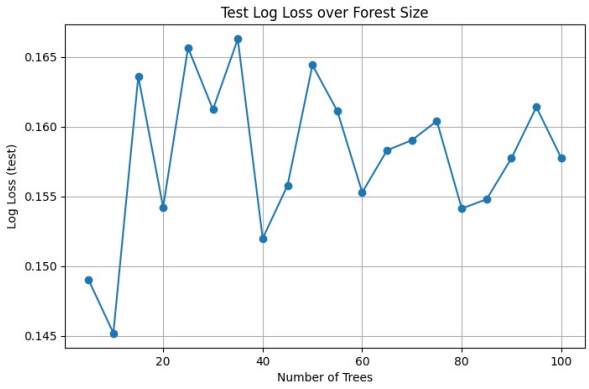


Fig. 6. Test log loss variation across forest size in Stage 3 pain intensity estimation.

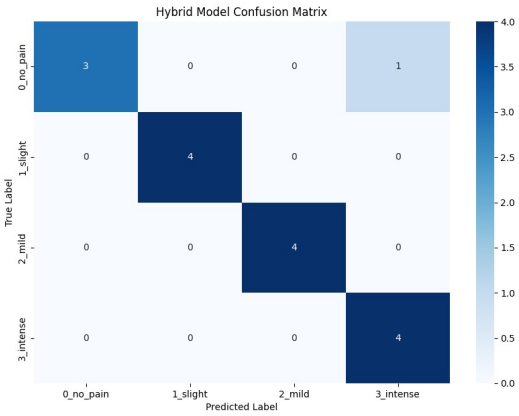


Fig. 7. Confusion matrix for Stage 3 pain intensity estimation (levels 0–3).

The Custom CNN comes out ahead, reflecting its stronger extraction of grayscale facial structure.

2) Stage 2: Pain Detection: Table III compares the two Stage 2 models; Fig. 4 and Fig. 5 show the confusion matrix and training curves for the MobileNetV2 variant.

MobileNetV2 is the better Stage 2 model on the strength of its perfect pain-class recall.

3) Stage 3: Pain Intensity Prediction: Table IV gives the per-class metrics for the hybrid Random Forest model, with its log loss behavior in Fig. 6 and confusion matrix in Fig. 7.

Overall accuracy reaches 93.75%, with a macro F1 of 0.94.

TABLE II
Per-class performance for Stage 1 (Custom CNN).

Class	Precision	Recall	F1-score
Angry	0.93	0.94	0.93
Happy	0.97	1.00	0.99
Neutral	0.94	0.91	0.93
Sad	0.91	0.90	0.90

TABLE III
STAGE 2 PAIN DETECTION RESULTS.

Model	Accuracy	Pain recall	No-pain precision
Custom CNN	52.00%	0.52	0.52
MobileNetV2	78.12%	1.00	1.00

TABLE IV
STAGE 3 PAIN INTENSITY CLASSIFICATION METRICS.

Class	Precision	Recall	F1
No pain	1.00	0.75	0.86
Slight	1.00	1.00	1.00
Mild	1.00	1.00	1.00
Intense	0.80	1.00	0.89

B. Results Discussion

Stage 1: The Custom CNN captures facial muscle activity more effectively than MobileNetV2. The main confusions occurred between neutral and sad, and between angry and neutral.

Stage 2: The Custom CNN overfit, showing high training accuracy but weak test performance, whereas MobileNetV2 generalized better and delivered higher pain recall.

Stage 3: The hybrid model produced stable multi-class predictions despite the limited dataset, with the slight and mild grades classified perfectly.

VI. Results and Discussion

This section reflects on overall system behavior, real-time performance, and limitations.

A. Real-Time Performance

- All models run in real time on GPU-equipped systems at 30–60 ms per frame.
- The Stage 1 and Stage 2 CNN models are light enough for edge deployment.
- The Stage 3 hybrid classifier infers nearly instantaneously.

B. Prediction Stability

- The Stage 1 CNN stays stable under head tilt and small lighting changes.
- Stage 2 MobileNetV2 detects pain consistently with minimal fluctuation.
- The Stage 3 hybrid classifier separates classes cleanly with little overlap.

C. System Behavior Across Emotions

- Happy is always filtered out at Stage 1, preventing false pain alerts.
- Angry, sad, and neutral can look visually similar, but the CNN separates them well.
- Pain predictions remain accurate under small head movements.

D. Limitations

- Small dataset size, particularly in Stage 3.
- Subject bias from repeated sampling.
- Sensitivity to extreme lighting conditions.
- Limited demographic diversity.

VII. Conclusion

This work presented a multi-stage facial expression and pain assessment framework intended for dependable real-time use in physical therapy and clinical monitoring. The system chains three levels: facial emotion recognition, pain detection, and pain intensity estimation. Experimentally, the Custom CNN proved highly accurate for emotion classification in Stage 1, MobileNetV2 delivered the most reliable pain-versus-no-pain detection in Stage 2, and the hybrid Random Forest model produced strong multi-class intensity predictions in Stage 3 despite the limited dataset. The architecture runs in real time, holds stable predictions across lighting variation and moderate head movement, and differentiates subtle pain-related facial cues effectively. Dataset scale and subject diversity remain constraints, but the results support the feasibility of lightweight, clinically aligned, and computationally efficient facial analysis systems. Future work will broaden the dataset, add demographic variety, incorporate temporal deep learning architectures, and evaluate the system in real clinical environments.

VIII. Future Work

Several extensions could further improve accuracy, scalability, and clinical value. First, growing the dataset to cover more subjects with varied facial structures, skin tones, and expression habits would strengthen generalization and help the classifier separate the subtle, person-specific patterns that pain produces. Second, the system could move beyond binary pain classification to recognize the seven universal expressions (happiness, sadness, anger, fear, surprise, disgust, and neutral), giving therapists a fuller emotional profile of the patient during therapy or clinical procedures.

Although landmark-based machine learning models are fast and efficient, future versions could adopt CNN or other deep architectures for fully automated analysis, removing the reliance on handcrafted landmarks; lightweight deep models in the spirit of compact FER networks [10], [14] could lift accuracy while still meeting real-time constraints. On the deployment side, wrapping the system in a Flask web service or an Android application would broaden access for therapists, hospitals, and remote-care settings. Finally, real-time alerting, for example notifications when pain levels rise abruptly, would make the system valuable in hospitals, rehabilitation centers, and assisted-care facilities where continuous monitoring underpins patient safety.

References

[1] R. Antel, S. Whitelaw, G. Gore, and P. Ingelmo, "Moving towards the use of artificial intelligence in pain management," *European Journal of Pain*, vol. 29, p. e4748, 2025.

[2] D. Ciraolo, M. Fazio, R. S. Calabrò, M. Villari, and A. Celesti, "Facial expression recognition based on emotional artificial intelligence for tele-rehabilitation," *Biomedical Signal Processing and Control*, vol. 92, p. 106096, 2024.

[3] M. Zhang *et al.*, "Using artificial intelligence to improve pain assessment and pain management: a scoping review," *Journal of the American Medical Informatics Association*, vol. 30, no. 3, pp. 570–587, 2023.

[4] S. Gwak and K. Park, "Designing effective visual feedback for facial rehabilitation exercises: investigating the role of shape, transparency, and age on user experience," *Healthcare*, vol. 11, no. 13, p. 1835, 2023.

[5] A. Celesti *et al.*, "Adopting machine learning-based pose estimation as digital biomarker in motor tele-rehabilitation," in *Proc. IEEE Symposium on Computers and Communications*, 2023.

[6] ——, "The tele-rehabilitation as a service (TRaaS) project: rationale, study design, and methodology," in *Proc. 2023 IEEE 19th International Conference on e-Science*, 2023.

[7] S. Sikka, S. Lamba, and A. Sharma, "Computer vision-based monitoring systems for home-based physiotherapy: a systematic review," *Archives of Physical Medicine and Rehabilitation*, 2023.

[8] B. Zou, X. Zhang, Y. Zhao, and R. Wei, "Deep learning for facial expression recognition: a review," *Pattern Recognition*, vol. 145, p. 109919, 2024.

[9] H. B. Ali and D. M. W. Powers, "Face and facial expression recognition fusion based non negative matrix factorization," in *Proc. ICAART International Conference on Agents and Artificial Intelligence*, 2015.

[10] H. Lu, R. Huang, C. Ma, C. Liu, and J. Feng, "A lightweight and high-speed deep neural network for facial expression recognition," *Applied Soft Computing*, vol. 103, p. 107142, 2021.

[11] S. Pini, S. Sani, L. Roffia, and D. Bovo, "Real-time pain assessment through video analysis of facial expressions: a survey," *Artificial Intelligence in Medicine*, vol. 124, p. 102244, 2022.

[12] T. M. Ashraf, M. M. Rahman, and M. G. Rabbani, "Automatic recognition of pain intensity using a deep spatio-temporal network on video sequences," *Journal of King Saud University – Computer and Information Sciences*, vol. 35, no. 6, p. 101533, 2023.

[13] M. Rasti and A. Moosavi-Mirkala, "Pain level prediction using facial landmark features and fusion of deep learning models," *Expert Systems with Applications*, vol. 239, p. 122245, 2024.

[14] Y. K. Al-Shayea, F. M. Ghaleb, and A. S. Hameed, "Automatic pain intensity estimation from facial expressions using a lightweight convolutional neural network," *Sensors*, vol. 21, no. 16, p. 5530, 2021.

[15] K. S. Kamalanand and A. P. Singh, "Automated pain assessment using facial micro-expressions with transfer learning and deep convolutional neural networks," *Biomedical Signal Processing and Control*, vol. 77, p. 103759, 2022.

[16] D. C. Cireşan, U. Meier, and J. Schmidhuber, "Multi-column deep neural networks for image classification," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2012, pp. 3642–3649.